Explainable AI in Learning Analytics: Improving Predictive Models and Advancing Transparency Trust

Qinyi Liu
Centre for the Science of Learning & Technology (SLATE)
University of Bergen
Bergen, Norway
qinyi.liu@uib.no

Mohammad Khalil
Centre for the Science of Learning & Technology (SLATE)
University of Bergen
Bergen, Norway
mohammad.khalil@uib.no

***Abstract*—As online education becomes more widely available, the amount of data available and accessible has exploded. Such a wealth of educational data provides ample opportunities for the fields of learning analytics and educational data mining to expand and yield numerous benefits, such as identifying at-risk students, providing personalized feedback, and providing actionable insights. Machine learning and deep learning techniques are commonly used for these and other purposes. However, simply applying these technologies is not enough without allowing humans to understand them. Because only those who understand the logic behind the technology can cultivate trust and make the technology work to its maximum effectiveness. To address the gap, this paper focuses specifically on a case study of explainable machine learning techniques for predicting student performance in online courses, and its contribution is twofold. First, we show how the use of explainable AI techniques can inform model diagnosis and direction forward in situations where performance is less than ideal. Secondly, we show how different types of explainable AI techniques can improve transparency and trust in educational scenarios, allowing stakeholders to benefit from explainable AI.**

Keywords—explainable learning analytics; transparency; trustworthiness; Artificial intelligence (AI); explainable AI

I. INTRODUCTION

Currently, due to the widespread availability of online education, more and more data is available. The availability of the immense amount of educational data has allowed the field of learning analytics and educational data mining to grow and bring many benefits. In the field of learning analytics and educational data mining, machine learning and deep learning techniques are widely used [17, 23]. Thanks to the use of increasingly sophisticated models, some predictions are able to achieve impressive results, such as achieving 80%-90% accuracy F1-score values in a prediction of whether a student will drop out or succeed [17]. Some models have achieved 91% accuracy in predicting student performance [8].

However, it is not enough to achieve high-performance predictions; in order to use the model and convince stakeholders, it is necessary not only to obtain good predictions but also to understand the reasons for such decisions. Stakeholders need to understand the logic and reasons behind the decisions. Especially in education and learning sciences, stakeholders such as students and teachers may be sceptical due to lack of good transparency, interpretability and unbiased evidence. Lack of trust and transparency can affect the further dissemination of Artificial intelligence (AI) technologies in education. In addition, the field of learning analytics, due to issues such as data quality/size [3], does not always have the utmost performance in terms of effectiveness of its predictive models. Even in such cases, a less high-accuracy performing model, if the model decision-making process can be explainable, it can help building user's trust in the model. In this context, Explainable Artificial Intelligence (XAI) may be one of the many means of addressing the above problem. XAI aims to enable human experts to understand the explanatory factors behind AI decisions [15]. The role of XAI is to provide model diagnostics in addition to enhancing model interpretability [15].

Based on the above background, the research questions of this paper are *1) How to apply different XAI techniques in learning analytics predictive models to enhance interpretability?* and *2) How to utilize XAI techniques for diagnosis of learning analytics predictive models in case of less high-accuracy model performance?*

II. RELATED WORK

A. Learning analytics predictive model

Learning analytics is a relatively young field, the recognized definition of learning analytics has been introduced as "the measurement, collection, analysis and reporting of data about learners and their contexts, for purposes of understanding and optimizing learning and the environments in which it occurs" [22]. And one of the most widely used methods among learning analytics is predictive modelling [27, 20] . Predictive modelling is a statistical technique used to determine the likelihood of something to happen. There are numerous places where predictive modelling is used in education, such as monitoring of student mental health issues, forecasting student dropout rates, and predicting student academic performance [4, 10]. Good prediction results can be very useful in identifying students in particular need of attention, reducing failure rates, and providing teachers with suggestions and reflections on how to teach effectively next semester [8, 10]. It can even influence educational decisions and resource allocation at the macro level across the school/university [10].

B. Explainable AI

AI has grown at an extremely high rate within the last decade and its adoption in academia and industry has been widespread. But a large part of this success has been achieved by complexed models, also referred to as black-box AI models. These models, for example neural networks, lack transparency to explain the outcomes of prediction models

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[30]. In that case, black box leads to increased uncertainty of models' operations and the final journey towards the decision. To address this problem, explainable artificial intelligence (XAI) has been proposed in the hope of enabling human experts to understand the underlying explanatory factors for AI decision making, enabling ethical responsible AI and transparent verifiable machine learning in the decision support [1]. Because of its inherent driving force, XAI has experienced swift evolution, leading to a development of numerous approaches and methods that enable explanation of machine learning models. The figure presented below, Fig. 1, illustrates the most widely adopted XAI techniques according to [22].

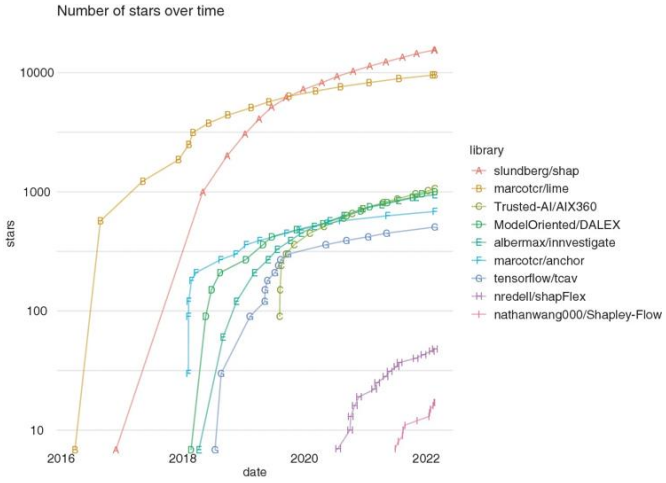


Fig. 1. The most popular XAI toolbox [5].

In terms of the categories of XAI, these methods can be divided into three types, global methods, which attempt to explain the entire model, local methods, which attempt to explain single predictions, and combining/enhancing methods providing explanation by means of aggregation or integration [7].

C. *Explainable AI in learning analytics*

There are already some examples of using XAI in education and learning, such as the previously mentioned scholars who performed student performance prediction followed by further analysis using XAI models [17], and scholars who demonstrated that using models with XAI features can be very effective in enhancing positive student perceptions of recommender systems [24]. In addition, the importance of applying XAI in learning analytics has been also highlighted (e.g., [18]). The cruciality of XAI in education can be encapsulated into two points: it acts as a powerful driver for enhancing understanding, trust, and accountability in learning analytics, while also supporting student-teacher interactions and AI literacy [9]. Furthermore, XAI's diagnostic capabilities are invaluable in scenarios where model performance is suboptimal, as it provides insights for identifying areas of improvement and guiding future directions [21].

III. METHODOLOGY

The main objective of this study is to provide a variety of different XAI methods for learning analytics algorithms to reveal the role of XAI in model diagnosis and trust enhancement in educational scenarios. Therefore, we followed the process shown in Fig. 2. The entire experimental pipeline of this model consists of three main sequential phases: (1) data handling & processing: selected data, cleaned data, normalised data. (2) data mining: applied selected classifiers to predict student performance (3) XAI: applied XAI methods according to characteristics of the model. The following figure visualises the exact flow of these steps (see Fig. 2).

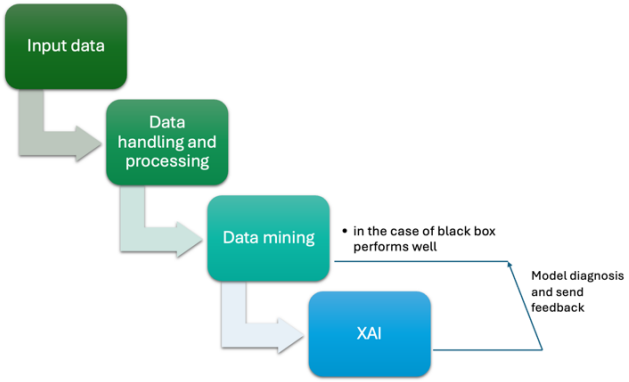


Fig. 2. Experiment pipeline.

A. *Data handling & processing*

Since the RQ2 problem of this paper lies in the commitment to exploring the role of XAI in model diagnosis when the learning analytics predictive model performs below optimal, a learning analytics dataset with some small data size and some missing values was chosen for the data selection in this paper. The MOOC dataset used in this paper is from a Norwegian Open edX platform with help of the learning analytics tool, OXALIC [19]. The data contained in this dataset consists of four main categories, namely course statistics (e.g. start and end times of individual courses, etc.), course interaction-related data (e.g. amount of forum interaction events), course video-related data (e.g. number of course videos, number of pauses), time spent on the platform (e.g. time spent on pages). This dataset currently contains data from only one course, which is aimed at students studying for a relevant master's degree. The details of the dataset pre-cleaning is demonstrated in Table 1.

TABLE I. DETAIL OF USED DATASET FOR XAI

Dataset characters	***Dataset detail***
Year	2021
Number of attributes	12
Number of records	64
Target variable	Continuous
Number of countinuous variables	12
Number of caregorical variables	0
Imblance ratio	0.64

TABLE II. VARIBALES USED IN THE DATASET FOR XAI

Variables	***Description***
studentEdxId	Student identification number
eventsCount	Number of total events by the students which includes all the activity by students.
sessionsCount	Number of sessions in a course.
sessionsDuration	Time spent in sessions

Variables	*Description*
problems_count	Number of problems in a session
problems_progress	Percentage metric showing progress of problems per course
problems_done	Number of problems done by students
attempts_count	Total number of attempts for problems by the student
attempts_accuracy	Accuracy of getting the correct answers to the problems
attempts_incorrect_count	Number of incorrect attempts
attempts_correct_count	Number of correct attempts
time_on_platform	Total time spent on the platform b y the students

The first stage of the pipeline, data handling & processing, was broken down into several steps. The first step is to select the data. Since the data set is scraped from the learning analytics tool OXALIC, the first step is to select all the learning behaviour records from the log data, such as the time spent by students on the platform. The second step is data cleaning. In this paper, we first checked the missing values. After checking, we found that the missing values are caused by students dropping courses. We handled such by examining missing target variable (i.e. students performance). Subsequently, we also removed some duplicate and overlap variables, for example, three variables "maximum time spent on the platform" "minimum time spent on the platform" and "average time spent on the platform" co-existed in the dataset. We remove two other overlapping variables besides "average time spent on the platform". The third step is data normalisation. In this paper, the target variables are converted to binary labels. As all students passed the pass line except for those in the drop course. Therefore, the criterion for conversion is the median performance grade. We transformed any performance greater than the median grade is "excellent" and the rest is "good". Finally, the machine learning pre-processing was prepared. Both datasets were partitioned 75:25 using the stratification method, and the training set was further partitioned using 10-fold cross-validation of the stratification.

B. Data mining

We use several algorithms that have performed well in the field of learning analytics and educational data mining, including support vector machines, random forests, KNN, and XGBoost. The first algorithm used support vector machines, a robust classification and regression technique that maximizes the predictive accuracy of a model without over-fitting the training data [13]. It works by mapping data to a high-dimensional feature space so that data points can be classified and then the features of the new data are used to predict which group the new record should belong to [13]. Support vector machines have also been widely used in the field of learning analytics and educational data mining, and some scholars have used support vector machines to predict STEM degree completion in community colleges over three years with good results [28]. In addition, some researchers in Nigeria have used support vector machines in an attempt to discover the relationship between students' pre-enrolment grades and their final academic performance, achieving an accuracy of 94%, outperforming other algorithms [12]. Based on these results, support vector machines have been well used to predict academic performance, so this paper uses support vector machines.

The second algorithm used is Random Forest. Random forest is a commonly used machine learning algorithm, registered by Leo Breiman and Adele Cutler, which combines the outputs of multiple decision trees to produce a single result. Its ease of use and flexibility has promoted its adoption, as it can handle both classification and regression problems [14]. The principle is to construct multiple decision trees to complete the task during training and to determine the outcome by evaluating the voting results of multiple independent decision trees [14]. Random forests have been widely used in the field of learning analytics [29]. For example, random forests have been used to predict whether students will earn an undergraduate degree [6], and 24 self-reported variables from questionnaires, including self-confidence and group learning, have been used to predict the performance of first-year college students [26]. All the former achieved good accuracy rates. Based on these results, random forests have been well used to predict academic performance, so this paper will also use random forests.

The third algorithm that has been used is KNN. KNN is defined as the classification of unknown data points based on nearest neighbours [11]. Specifically, the nearest neighbours are calculated based on the k-value, which determines the number of nearest neighbours to be considered, thus defining the class of a sample data point [11]. KNN has achieved good results in predicting student performance. For example, Bhardwaj and Pal [25] have used KNN, decision trees and Bayes classifier to predict student performance at university, where the KNN results showed good performance results, better than Bayes classifier.

The fourth algorithm being used is XGBoost. XGBoost stands for eXtreme Gradient Boosting. xGBoost provides a parallel tree boosting (also known as GBDT, GBM) to solve many data science problems in a fast and accurate manner [16]. xGBoost achieved the best results among multiple machine learning algorithms on some educational datasets [16]. Finally, it is added that white-box algorithms such as logistic regression were also tried in this research, as they also obtained good performance on some predictions of student performance considering any. However, they performed lower on our dataset, for example logistic regression had an accuracy of 50%. The results of the accuracy of each algorithm are shown in the table below (see Table III). In order to obtain a more stable and reliable performance evaluation, we train and test each model five times during the experiment and calculate the average of accuracy, AUC and F1-score.

TABLE III. AVERAGED PERFORMANCE RESULTS OF ALGORITHMS

Machine learning algorithm	*Accuracy*	*AUC*	*F1-Score*
Support vector machine	0.44	0.46	0.57
Random forest	0.74	0.83	0.74
KNN	0.46	0.44	0.54
XGBoost	0.71	0.73	0.67
Logic regression	0.50	0.62	0.59

From the above table, it is clear that random forest performs the best but the performance is not particular high when compare to other performance prediction example [8,

17]. Therefore, random forest is selected as the next algorithm for explainable machine learning analytics for further explainability enchantment and model diagnose.

C. XAI techniques

As for specific XAI used in this learning analytics case, both global methods and local methods has been applied. XAI because the stakeholders in the education domain need to have both a model grasp and an interpretation of individual forecasts. In the global methods section, we use Partial dependence plot (PDP) and Accumulated local effects (ALE). PDP is a global XAI method. The reason for choosing PDP is to show the marginal effect or contribution of individual features to the predictive value of the black-box model [5], this information can clearly inform how much value students and teachers have in which aspect of their efforts. In addition, among global methods, we also use ALE. The ALE plot highlights the impact of specific features on the predictions of the machine learning model by partially isolating the impact of other features [5]. The resulting ALE interpretation centres on the average effect of the features such that the main feature effect is relative to the average prediction of the data [5]. The reason for choosing ALE is because it is unbiased and remains valid even when features are correlated [5]. This can complement the PDP. Additionally, the utilization of feature importance in model diagnosis is crucial as it offers insights into which features are most influential in driving the model's predictions, thereby guiding further analysis and potential improvements [2].

As for local methods, we used LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations). LIME is a technique that approximates any black-box machine learning model with a locally interpretable model to explain each individual prediction. The rationale behind it is to test what happens to a prediction when a machine learning model is fed with different data in order to understand why the machine learning model makes a certain prediction [5]. The advantage of LIME is that it can give explanations of individual predictions that can be easily understood by non-specialist backgrounds and is applicable to a wide range of data [5]. Given the power of LIME in enhancing the interpretability of individual predictions, and its friendliness to people without technical backgrounds, we decided to use it to add explanatory features to individual student performance predictions. SHAP (SHapley Additive exPlanations) is chosen for its practicality and specific advantages in the educational context. It provides a coherent and consistent measure of feature importance based on game theory, which can be particularly useful in complex models where understanding individual feature contributions is crucial. SHAP values not only reveal the impact of features on predictions but also how they interact with each other, offering a more nuanced understanding than global methods alone [5]. From the standpoint of the educational context, SHAP helps stakeholders like teachers and students understand the exact contribution of factors such as study duration to their performance predictions.

IV. Results and Discussion

A. Global methods results

In the context of global methods XAI (or model-level XAI), the Partial Dependence Plot (PDP) and Accumulated Local Effects (ALE) provide a comprehensive understanding of feature influences globally. The PDP for 'attempts_count' shows a threshold beyond which the prediction stabilizes. Understanding this can guide model refinement by focusing on the data range before the plateau. (see detail in Fig 3). Teachers can use this information to identify a possible 'sweet spot' of attempts after which student performance does not significantly improve, helping to optimize study plans.

ALE offers good insights in terms of model improvement and insights for education stakeholders (see detail in Figure 4). In terms of model improvement, the ALE plot shows a smaller effect of 'sessionsDuration' on predictions, which suggests examining this feature's interaction with others could refine model understanding. As for insights to stakeholders, Fig. 4 shows that after around 10K session time, the performance increase is not significant any more. In the education context, this indicating a potential threshold for effective study time, which could guide the design of educational sessions focused on quality rather than duration.

B. Local methods results

For local-level XAI, LIME and SHAP offer granular insights. The LIME explanation helps in understanding the model's prediction for an individual instance by breaking down feature contributions. In education context, for a given student, teachers can see which factors are influencing their predicted performance most, allowing for more personalized feedback and intervention. In the Fig 5, we chose a random student's LIME result to interpret.

The prediction probabilities indicate that the model predicts the student's performance as 'good' with a high probability of 0.92 versus 'excellent' with a probability of 0.08. The features influencing this prediction are laid out in descending order of impact. A feature, attempts_count, with a value of 57, significantly decreases the likelihood of an 'excellent' prediction by contributing negatively (shown in red), suggesting a high number of attempts is linked to lower performance. Other features such as sessionsDuration, eventsCount, time_on_platform, and sessionsCount have varying degrees of positive (blue) and negative (red) impact on the 'excellent' prediction probability. These values offer a nuanced look at what the model considers when making predictions, allowing educators to understand the model's decision-making process at an individual level.

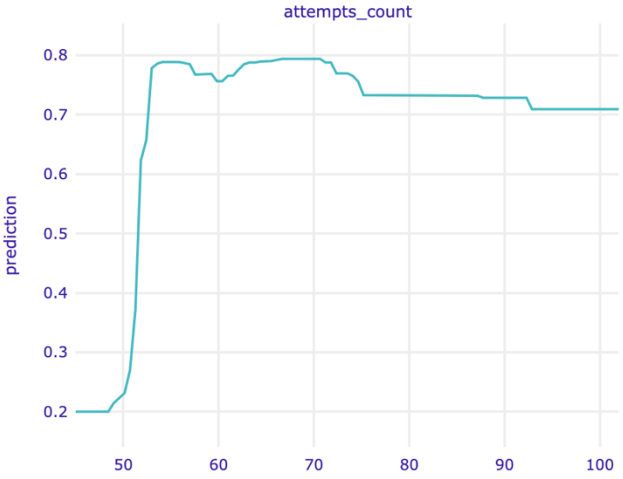


Fig. 3. Partial Dependence Plot (PDP) for Random Forest Model.

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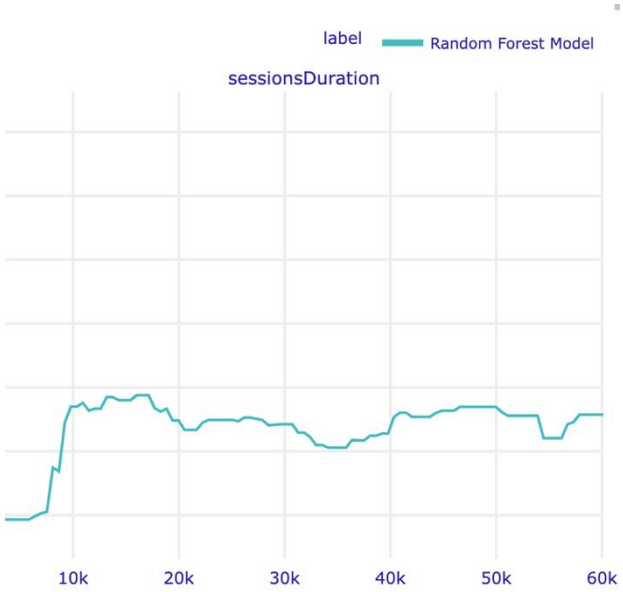


Fig. 4. Partial Dependence Plot (PDP) for Random Forest Model.

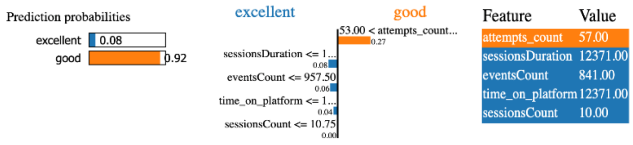


Fig. 5. LIME reulst of predictive model

SHAP values indicate the contribution of each feature to the prediction, offering a detailed understanding that can be used for model debugging and improvement. The detail of SHAP value can be seen from Fig.6 below. The `attempts_count` with a value of 102 pushes the prediction lower by a significant amount (red bar), indicating that a higher number of attempts may be associated with a lower performance prediction. Conversely, features such as `sessionsDuration`, `time_on_platform`, and `sessionsCount` are pushing the prediction higher (blue bars), suggesting that these factors contribute positively to the student's predicted performance. Each feature's SHAP value shows its contribution to the difference between the base value and the actual model output. The sum of all SHAP values plus the base value gives us the prediction for that particular student. The visualization helps to dissect and understand the model's decision on a granular level, which can be invaluable for educational interventions tailored to individual students' needs.

Additionally, an intriguing observation emerges when comparing the findings from Figure 6 to the PDP values, highlighting discernible differences between them. For PDP values, "attempts_counts" representing the number of attempts has some downward trend as the number of attempts increases well before 100. This is different from the 102 times shown in Fig. 6 that have a negative impact on the results. The reason may be that PDP shows the average impact of a certain feature on the prediction results of the entire dataset, which can sometimes be misleading if there is a significant interaction between features [16]. Therefore, the difference in the values of these two XAI metrics also provides useful suggestions for model diagnosis: check for interactions.

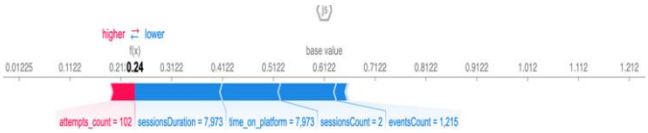


Fig. 6. SHAP vaules of predictive model

Through the discussion above, we can integrate and conclude in relation to the research questions posed. For RQ1, on applying different XAI techniques in learning analytics predictive models to enhance interpretability, it's evident that both global (e.g., PDP and ALE) and local (e.g., LIME and SHAP) methods enable an understanding of feature impacts at macro and micro levels. Global methods like PDP and ALE reveal the overarching influence of specific features such as "attempts_count" and "sessionsDuration" on model predictions, guiding model refinement and the design of educational interventions. Local methods like LIME and SHAP provide individual-level explanations, allowing educators to offer personalized feedback and interventions based on a model's prediction for a single student. For RQ2, concerning the utilization of XAI techniques for diagnosing learning analytics predictive models in case of less high-accuracy model performance, the analysis of LIME and SHAP results uncovers the differential impacts of features on predictions. This variation provides clues for model diagnosis. For instance, a high attempts_count negatively impacting performance prediction in SHAP analysis, compared to its average impact across the dataset as shown by PDP, suggests the need to further examine feature interactions for diagnosing and improving the model.

C. *Implications*

The results we retrieved from our study suggest that XAI can provide good model diagnostics for learning analytics prediction models and better explain the inner logic of the algorithm, increasing the interpretability and transpracy towards the algorithm. Enhancing the interpretability and transparency of the algorithm is paramount in fostering trust among relevant stakeholders. In a specific educational context, this implies that students and teachers may be more inclined to trust or accept the results of predictive algorithms due to the presence of Explainable Artificial Intelligence (XAI). This facilitates the integration of learning analytics, such as predictive algorithms, into the educational process. Furthermore, students and educators can not only use the model's predictive outcomes to understand which behaviors might lead to better or worse academic performance but can also employ XAI to identify personalized learning strategies. By tailoring interventions to individual needs and deepening the understanding of the learning process itself, this approach can provide a more informed decision-making process for curriculum design, teaching strategies, and student engagement techniques, ultimately aiming to enhance educational outcomes and personalize the learning experience.

V. CONCLUSIONS AND FUTURE DIRECTION

This paper uses a small sample of MOOC dataset to explore the significant role of different XAI techniques in learning analytics predictive modelling direction in two aspects, namely model diagnosis, enhanced interpretability

and transparency. The contributions of this paper are: first, considering the problem that data quality/size in the learning analytics domain are often limiting and ultimately lead to suboptimal accuracy, this paper demonstrates the important role of XAI techniques in model diagnostics through an example of a learning analytics predictive model with a not-very-high-accuracy (<80%) in education domain. This further lays the groundwork for subsequent model enhancement. Second, it demonstrates the role of XAI in increasing stakeholders's understanding of and trust in the model. Stakeholders can proactively use the insights revealed by the XAI technique to make changes in their own behaviour in addition to accepting the predicted results . For example, an individual student can use the local XAI results to see which of his or her behaviours the algorithm determines to be important for better result, and thus target that behaviours for improve grades.

As for the limitation and the future direction, the current study showcases the potential of XAI within a learning analytics context using a singular MOOC dataset. Future research might expand the breadth of tested data, incorporating a wider array of MOOC environments to validate the robustness of XAI interpretations. Moreover, while various XAI methods were explored in this field, XAI still has many methods that can be further applied in the education data science domain. Additionally, future studies require investigating the reception of XAI feedback by stakeholders. Subsequent investigations could include interviews or surveys post-XAI result presentation to gauge stakeholder engagement and the practical utility of XAI in educational settings. This feedback will be vital in understanding the real-world impact of XAI, its effectiveness in improving educational outcomes, and how it shapes the behaviours and trust of educators and learners in learning analytics and educational data mining.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

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